



Gnanamani College of Technology

(AUTONOMOUS)

Approved by AICTE, New Delhi & Affiliated to Anna University, Chennai
NH-7, A.K. Samuthiram, Pachal – 637018, Namakkal (Dt.), Tamilnadu, India.
Web : www.gct.org.in, E-Mail : info@gct.org.in, Contact No. 7598293888, 7598293999

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

23CSE24 - SOFTWARE TESTING AND

AUTOMATION

LAB MANUAL

ACADEMIC YEAR: 2026-2027(ODD SEMESTER)

LIST OF EXPERIMENTS:

1. Develop the test plan for testing an e-commerce web/mobile application (www.amazon.in).
2. Design the test cases for testing the e-commerce application
3. Test the e-commerce application and report the defects in it.
4. Develop the test plan and design the test cases for an inventory control system.
5. Execute the test cases against a client server or desktop application and identify the defects.
6. Test the performance of the e-commerce application.
7. Automate the testing of e-commerce applications using Selenium.
8. Integrate TestNG with the above test automation.
9. Mini Project:
 - a) Build a data-driven framework using Selenium and TestNG
 - b) Build Page object Model using Selenium and TestNG
 - c) Build BDD framework with Selenium, TestNG and Cucumber

Exp. No: 1	Develop the Test Plan for Testing an E-Commerce Web/Mobile Application – Amazon.in
Date:	

Aim:

To develop test cases for testing the major functions of the Amazon.in e-commerce application.

Objective:

- To understand e-commerce application testing.
- To identify important test scenarios.
- To prepare test cases for major functions.
- To verify that the application works as expected.

Prerequisite Knowledge:

Basic knowledge of:

- Software testing
- Test cases
- Web/mobile applications
- Functional testing
- SDLC

Theory / Concepts:

A **test case** is a set of steps, input data, and expected results used to check whether a particular feature works correctly.

Important testing concepts used in this experiment are:

- Functional testing
- Black-box testing
- Positive testing
- Negative testing
- Regression testing
- Usability testing

Advantages:

1. Finds defects early.
2. Improves software quality.
3. Provides systematic testing.
4. Test cases can be reused.

5. Helps ensure customer satisfaction.

Disadvantages:

1. Preparing test cases takes time.
2. Large applications require many test cases.
3. Test cases must be updated when requirements change.
4. Some real-world conditions are difficult to test.

Algorithm:

1. Open Amazon.in.
2. Identify the major application modules.
3. Select a function to test.
4. Prepare test input.
5. Define the expected result.
6. Execute the test case.
7. Compare actual and expected results.
8. Mark the test case as Pass or Fail.
9. Record defects if any.
10. Complete the test report.

Application:

This testing approach can be used for:

- Amazon
- Flipkart
- Myntra
- Online grocery applications
- Online ticket-booking applications
- Other e-commerce websites and mobile apps

Real-Time Example:

A customer searches for a mobile phone, selects a product, adds it to the cart, and proceeds to checkout. Test cases are prepared to verify each step of this shopping process.

TEST CASES

TC ID	TEST CASE	TEST DATA	EXPECTED RESULT	STATUS
TC01	Open website	Amazon.in	Home page should open	Pass
TC02	Login with valid details	Valid credentials	User should login successfully	Pass
TC03	Login with invalid details	Wrong password	Error message should display	Pass
TC04	Search product	“Mobile Phone”	Relevant products should display	Pass
TC05	Search invalid product	“XYZ123”	No-result message should display	Pass
TC06	Select product	Click product	Product details should display	Pass
TC07	Add product to cart	Select product	Product should be added to cart	Pass
TC08	Remove product from cart	Click Remove	Product should be removed	Pass
TC09	Change quantity	Quantity = 2	Quantity and price should update	Pass
TC10	Add product to wishlist	Select product	Product should be added to wishlist	Pass
TC11	Apply filter	Price/category filter	Matching products should display	Pass

TC12	Sort products	Price low to high	Products should be sorted correctly	Pass
TC13	Add address	Valid address	Address should be saved	Pass
TC14	Checkout	Product in cart	Checkout page should open	Pass
TC15	Select payment method	Test payment option	Payment option should be selected	Pass
TC16	Successful payment	Authorized test data	Order should be placed	Pass
TC17	Failed payment	Invalid/test failure	Proper error should display	Pass
TC18	Order history	Existing order	Order should be displayed	Pass
TC19	Cancel order	Eligible order	Order should be cancelled	Pass
TC20	Return product	Eligible product	Return request should be created	Pass
TC21	Logout	Click Logout	User should be logged out	Pass

Note: Payment test cases should be executed only with an authorized test/sandbox environment, not with real payment credentials.

Sample Input

Username: Test User
Search: Mobile Phone
Quantity: 1

Sample Output

Login Successful
Product Search Successful
Product Added to Cart
Checkout Successful
Test Case: PASS

Viva Questions with Answers

1. What is a test case?

A test case is a set of steps and conditions used to verify a software feature.

2. What is software testing?

Software testing is the process of finding defects and verifying that software meets its requirements.

3. What type of testing is used in this experiment?

Functional and black-box testing.

4. Why is e-commerce testing important?

It ensures that important functions such as search, cart, payment, and orders work correctly.

5. What is positive testing?

Testing the application with valid input.

6. What is negative testing?

Testing the application with invalid or unexpected input.

7. What is regression testing?

Testing existing functionality again after changes are made to the application.

8. What is an expected result?

The result that should occur according to the requirements.

9. What is an actual result?

The result obtained when the test case is executed.

10. What is a failed test case?

A test case is failed when the actual result does not match the expected result.

Result:

Thus, the test plan and test cases for the Amazon.in e-commerce web/mobile application were prepared successfully.

Exp. No: 2	Design the Test Cases for Testing the E-Commerce Application
Date:	

Aim:

To design test cases for testing the major functions of an e-commerce application such as Amazon.in.

Objective:

- To identify important e-commerce functions.
- To design suitable test cases.
- To verify the functionality of the application.
- To identify defects using test-case execution.

Prerequisite Knowledge:

Basic knowledge of:

- Software testing
- Test cases
- Functional testing
- Web applications
- SDLC

Theory:

A **test case** is a set of conditions, inputs, steps, and expected results used to verify a particular function of an application.

In an e-commerce application, important functions include login, product search, product details, cart, checkout, payment, and order management.

Advantages:

1. Helps identify defects.
2. Provides systematic testing.
3. Improves software quality.
4. Test cases can be reused.

5. Helps verify application requirements.

Disadvantages:

1. Preparing test cases takes time.
2. Large applications require many test cases.
3. Test cases need modification when requirements change.

Algorithm:

1. Identify the main functions of the e-commerce application.
2. Select the function to be tested.
3. Define the test input.
4. Write the test steps.
5. Define the expected result.
6. Execute the test case.
7. Compare actual and expected results.
8. Mark the test case as Pass or Fail.
9. Record any defects.
10. Complete the test report.

TEST CASES

TC ID	Test Case	Test Input	Expected Result	Status
TC01	Open application	Amazon.in	Home page should open	Pass
TC02	Login with valid details	Valid username/password	User should login successfully	Pass
TC03	Login with invalid details	Wrong password	Error message should appear	Pass
TC04	Search for a product	"Mobile Phone"	Relevant products should be displayed	Pass

TC05	Search with invalid keyword	"XYZ123"	Appropriate no-result message should appear	Pass
TC06	Select a product	Click a product	Product details should be displayed	Pass
TC07	Add product to cart	Select product	Product should be added to cart	Pass
TC08	Remove product from cart	Click Remove	Product should be removed	Pass
TC09	Change product quantity	Quantity = 2	Quantity and total price should update	Pass
TC10	Apply product filter	Price/category filter	Matching products should be displayed	Pass
TC11	Sort products	Price: Low to High	Products should be sorted correctly	Pass
TC12	Add product to wishlist	Select product	Product should be added to wishlist	Pass
TC13	Add delivery address	Valid address	Address should be saved	Pass
TC14	Proceed to checkout	Product in cart	Checkout page should open	Pass
TC15	Select payment method	Valid test payment option	Payment method should be selected	Pass
TC16	Complete order	Authorized test data	Order should be placed successfully	Pass
TC17	Failed payment	Test failure condition	Proper error message should appear	Pass

TC18	View order history	Existing order	Order should be displayed	Pass
TC19	Cancel order	Eligible order	Order should be cancelled	Pass
TC20	Return product	Eligible product	Return request should be created	Pass
TC21	Logout	Click Logout	User should be logged out	Pass

Sample Test Case:

Test Case ID: TC04

Test Case: Search for a product

Input: Mobile Phone

Steps:

1. Open the e-commerce application.
2. Enter "Mobile Phone" in the search box.
3. Click the Search button.

Expected Result: Relevant mobile-phone products should be displayed.

Actual Result: Relevant mobile-phone products are displayed.

Status: PASS

Sample Input:

Search Product: Mobile Phone

Quantity: 1

Sample Output:

Product search completed successfully.

Relevant products displayed.

Test Case Status: PASS

Application

The test cases can be used for:

- Amazon
- Flipkart
- Myntra
- Meesho
- Other e-commerce web and mobile applications

Viva Questions with Answers

1. What is a test case?

A test case is a set of steps and conditions used to verify a software function.

2. What is the purpose of a test case?

To verify whether a particular functionality works as expected.

3. What is functional testing?

Functional testing checks whether the software functions according to its requirements.

4. What is expected result?

It is the result that should be produced by the application.

5. What is actual result?

It is the result obtained after executing the test case.

6. What is PASS?

A test case is PASS when the actual result matches the expected result.

7. What is FAIL?

A test case is FAIL when the actual result does not match the expected result.

8. Why do we test an e-commerce application?

To ensure that functions such as login, search, cart, checkout, payment, and orders work correctly.

Result:

Thus, the test cases for testing the e-commerce application were successfully designed and documented.

Exp. No: 3	Test the E-Commerce Application and Report the Defects in It
Date:	

Aim:

To test an e-commerce application and identify defects present in its functionality.

Objective:

- To execute test cases on an e-commerce application.
- To identify and record defects.
- To prepare a defect report.
- To improve software quality through defect tracking.

Prerequisite Knowledge:

- Software Testing Basics
- Test Cases
- Functional Testing
- Defect Life Cycle
- Web Applications

Theory:

A **defect (bug)** is a flaw in software that causes the application to produce incorrect or unexpected results. Defect reporting helps developers identify and fix issues before software release.

Advantages:

1. Improves software quality.
2. Identifies functional issues.
3. Enhances user experience.
4. Reduces business risks.

Disadvantages:

1. Time-consuming process.
2. Requires repeated testing.

3. Large applications may contain many defects.

Algorithm:

1. Open the e-commerce application.
2. Execute the prepared test cases.
3. Compare actual results with expected results.
4. Identify defects.
5. Record defect details.
6. Assign severity and priority.
7. Prepare the defect report.
8. Submit the report.

Test Execution Report:

TC ID	TEST SCENARIO	EXPECTED RESULT	ACTUAL RESULT	STATUS
TC01	Open Website	Home page opens	Home page opened	PASS
TC02	Login with valid credentials	Login successful	Login successful	PASS
TC03	Search Product	Products displayed	Products displayed	PASS
TC04	Add Product to Cart	Product added	Product added	PASS
TC05	Update Quantity	Quantity updated	Quantity updated	PASS
TC06	Remove Product	Product removed	Product removed	PASS
TC07	Checkout	Checkout page opens	Checkout page opens	PASS
TC08	Invalid Login	Error message displayed	No error message displayed	FAIL

TC09	Empty Search	Validation message displayed	Blank page displayed	FAIL
TC10	Invalid Payment	Error message displayed	Application freezes	FAIL

Defect Report

Defect ID	Module	Defect Description	Severity	Priority	Status
BUG01	Login	No error message for invalid login	Medium	High	Open
BUG02	Search	Blank page displayed for empty search	Medium	Medium	Open
BUG03	Payment	Application freezes during invalid payment attempt	High	High	Open

Sample Defect Description

Defect ID: BUG01

Module: Login

Steps to Reproduce:

1. Open the application.
2. Enter a valid username.
3. Enter an invalid password.
4. Click Login.

Expected Result:

An error message should be displayed.

Actual Result:

No error message is displayed.

Severity: Medium

Priority: High

Status: Open.

Application:

This defect-reporting process can be used for:

- Amazon
- Flipkart
- Myntra
- Meesho
- Any e-commerce website or mobile application

Viva Questions with Answers**1. What is a defect?**

A defect is an error or flaw in software.

2. What is defect reporting?

Defect reporting is the process of documenting software defects.

3. What is severity?

Severity indicates the impact of a defect on the application.

4. What is priority?

Priority indicates how quickly a defect should be fixed.

5. What is a defect life cycle?

It is the process through which a defect moves from identification to closure.

6. What is a failed test case?

A test case fails when the actual result differs from the expected result.

7. What is regression testing?

Retesting existing functionality after defect fixes.

8. Why is defect tracking important?

It helps ensure defects are fixed and software quality is maintained.

Result:

Thus, the e-commerce application was tested, defects were identified, and the defect report was prepared successfully.

Exp. No:4 Date:	Develop the Test Plan and Design the Test Cases for an Inventory Control System
----------------------------------	--

Aim:

To develop a test plan and design test cases for testing an Inventory Control System.

Objective

- To understand the functions of an inventory system.
- To prepare a simple test plan.
- To design test cases for inventory operations.
- To identify whether the system works as expected.

Prerequisite Knowledge:

Basic knowledge of:

- Software testing
- Test cases
 - Functional testing
- Database concepts
- Inventory management

Theory:

An **Inventory Control System** is used to manage products, stock quantities, suppliers, purchases, sales, and stock availability.

Testing verifies functions such as:

- User login
- Adding products
- Updating stock
- Searching products
- Purchasing stock

- Selling products
- Low-stock alerts
- Generating reports

Test Plan

Item	Description
Application	Inventory Control System
Testing Type	Functional Testing
Scope	Login, products, stock, suppliers, sales and reports
Test Environment	Web/Desktop application
Test Data	Product, supplier and stock details
Expected Result	System should work according to requirements

Advantages:

1. Prevents incorrect stock records.
2. Improves inventory accuracy.
3. Helps identify defects early.
4. Reduces manual errors.
5. Improves stock management.

Disadvantages:

1. Testing requires time.
2. Large inventory systems require many test cases.
3. Test data must be maintained.
4. Changes in requirements require retesting.

Algorithm:

1. Start the Inventory Control System.
2. Login with valid credentials.
3. Add and update product details.
4. Check product stock quantity.
5. Test purchase and sales operations.
6. Verify stock quantity after each transaction.
7. Test low-stock notification.
8. Generate inventory reports.
9. Compare actual and expected results.
10. Record Pass/Fail.
11. Stop.

TEST CASES

TC ID	TEST CASE	TEST INPUT	EXPECTED RESULT	STATUS
TC01	Login with valid details	Valid username/password	User should login successfully	Pass
TC02	Login with invalid details	Wrong password	Error message should appear	Pass
TC03	Add new product	Product name, ID, price	Product should be added	Pass
TC04	Add duplicate product	Existing product ID	Duplicate product should not be added	Pass
TC05	Update product	Change product price	Updated price should be saved	Pass

TC06	Search product	Product name/ID	Correct product should be displayed	Pass
TC07	Add stock	Quantity = 50	Stock should increase by 50	Pass
TC08	Remove stock	Quantity = 10	Stock should decrease by 10	Pass
TC09	Enter negative stock	Quantity = -5	Invalid quantity should be rejected	Pass
TC10	Sell product	Quantity = 5	Stock should decrease by 5	Pass
TC11	Sell unavailable product	Quantity greater than stock	Sale should be rejected	Pass
TC12	Low-stock check	Stock below limit	Low-stock alert should appear	Pass
TC13	Add supplier	Valid supplier details	Supplier should be added	Pass
TC14	Delete product	Select product	Product should be deleted	Pass
TC15	Generate stock report	Select report	Correct stock report should be displayed	Pass
TC16	Logout	Click Logout	User should be logged out	Pass

Sample Test Case:

Test Case ID: TC07

Test Case: Add stock

Input:

Product: Keyboard

Existing Stock: 20

New Stock: 50

Expected Result:

Total stock should become **70**.

Actual Result:

Total stock becomes **70**.

Status: PASS

Sample Input:

Product Name: Keyboard

Product ID: P101

Existing Stock: 20

Added Stock: 50

Sample Output:

Product added successfully.

Updated Stock: 70

Test Case Status: PASS

Application:

Inventory control systems are used in:

- Supermarkets
- Warehouses
- Retail shops
- Manufacturing companies
- Pharmacies
- E-commerce businesses

Viva Questions with Answers

1. What is an Inventory Control System?

It is a system used to manage and track products and stock quantities.

2. What is stock?

Stock is the quantity of products available in inventory.

3. What is a test case?

A test case contains inputs, steps, and expected results used to verify a function.

4. Why is inventory testing important?

It ensures that stock quantities and inventory transactions are accurate.

5. What happens when stock becomes low?

The system should generate a low-stock alert according to the configured limit.

6. What is functional testing?

It verifies whether the system functions according to its requirements.

7. What is a duplicate product?

A product that already exists in the inventory with the same identifying information.

8. What should happen when a user tries to sell more products than available?

The system should reject the transaction and display an appropriate message.

Result:

Thus, the **test plan was developed and test cases were successfully designed for the Inventory Control System.**

Exp. No: 5 Date:	Execute the Test Cases Against a Client-Server or Desktop Application and Identify the Defects
-----------------------------------	---

Aim:

To execute test cases on a client-server or desktop application and identify and report defects.

Objective:

- To execute prepared test cases.
- To compare expected and actual results.
- To identify defects.
- To prepare a defect report.

Prerequisite Knowledge

- Software testing
- Test cases
- Functional testing
- Defect life cycle
- Client-server and desktop applications

Theory:

Test execution is the process of running test cases and comparing the actual result with the expected result.

A **defect** is an error or problem in the application where the actual result does not match the expected result.

Advantages:

1. Helps find defects.
2. Improves software quality.
3. Verifies application functionality.
4. Helps developers correct errors.

Disadvantages:

1. Test execution requires time.
2. Large applications require many test cases.
3. Repeated testing may be required after defect fixes.

Algorithm:

1. Start the application.
2. Select the test cases.
3. Enter the required test data.
4. Execute each test case.
5. Record the actual result.
6. Compare actual and expected results.
7. Mark the test case as Pass or Fail.
8. Record failed cases as defects.
9. Assign severity and priority.
10. Prepare the defect report.
11. Stop.

Test Case Execution

TCID	TEST CASE	EXPECTED RESULT	ACTUAL RESULT	STATUS
TC01	Open application	Application opens	Application opened	PASS
TC02	Valid login	Login successful	Login successful	PASS
TC03	Invalid login	Error displayed	Error displayed	PASS
TC04	Empty login fields	Validation displayed	Validation displayed	PASS
TC05	Add product	Product added	Product added	PASS

TC06	Duplicate product	Duplicate rejected	Duplicate accepted	FAIL
TC07	Update product	Product updated	Product updated	PASS
TC08	Delete product	Product deleted	Product deleted	PASS
TC09	Search product	Product displayed	Product displayed	PASS
TC10	Search invalid product	No result displayed	No result displayed	PASS
TC11	Add stock	Stock increased	Stock increased	PASS
TC12	Remove stock	Stock decreased	Stock decreased	PASS
TC13	Negative quantity	Value rejected	Value accepted	FAIL
TC14	Zero quantity	Validation displayed	Validation displayed	PASS
TC15	Sell product	Stock decreases	Stock decreases	PASS
TC16	Sell excess stock	Transaction rejected	Transaction rejected	PASS
TC17	Low-stock alert	Alert displayed	Alert displayed	PASS
TC18	Add supplier	Supplier added	Supplier added	PASS
TC19	Delete supplier	Supplier deleted	Supplier deleted	PASS
TC20	Purchase stock	Stock increases	Stock increases	PASS
TC21	Generate stock report	Report displayed	Report displayed	PASS
TC22	Generate sales report	Report displayed	Report displayed	PASS
TC23	Update supplier	Supplier updated	Supplier updated	PASS

TC24	Logout	User logged out	User logged out	PASS
TC25	Close application	Application closes	Application closes	PASS

Defect Report

Defect ID	Test Case	Defect	Severity	Priority	Status
BUG01	TC06	Deleted product is still displayed	Medium	High	Open
BUG02	TC07	System accepts negative stock quantity	High	High	Open

Defect Details

BUG01 – Deleted Product Still Displayed

- **Module:** Product Management
- **Steps:** Select product → Click Delete → Search for product
- **Expected:** Product should not be displayed.
- **Actual:** Deleted product is still displayed.
- **Severity:** Medium
- **Priority:** High

BUG02 – Negative Quantity Accepted

- **Module:** Inventory Management
- **Steps:** Enter -10 as stock quantity → Click Save
- **Expected:** System should reject the negative value.
- **Actual:** System accepts the negative value.
- **Severity:** High
- **Priority:** High

Sample Input:

Product ID: P101

Product Name: Keyboard

Quantity: -10

Sample Output:

Test Case Execution Completed

TC01 – PASS

TC02 – PASS

TC03 – PASS

TC04 – PASS

TC05 – PASS

TC06 – FAIL

TC07 – FAIL

TC08 – PASS

Defects Identified: 2

Application:

This procedure can be used for:

- Inventory management systems
- Banking applications
- Payroll systems
- Library management systems
- Hospital management systems
- Client-server applications
- Desktop applications

Viva Questions with Answers

1. What is test execution?

Test execution is the process of running test cases and recording their results.

2. What is a defect?

A defect is a problem where the actual result differs from the expected result.

3. What is a failed test case?

A test case fails when its actual result does not match the expected result.

4. What is severity?

Severity indicates the impact of a defect on the application.

5. What is priority?

Priority indicates how urgently a defect should be fixed.

6. What is a defect report?

A defect report records details of an identified defect, such as its description, severity, priority, and status.

7. What is regression testing?

Regression testing verifies that existing functionality still works after changes or defect fixes.

Result:

Thus, the test cases were successfully executed against the application, and the identified defects were recorded in the defect report.

Ex. No:6	Test the Performance of an E-Commerce Application
Date:	

Aim:

To test the performance of an e-commerce application under different user loads.

Objective:

- To measure the response time of the application.
- To check the application under multiple users.
- To identify performance issues.
- To check the reliability of the application.

Prerequisite Knowledge:

- Basic knowledge of software testing.
- Basic knowledge of web applications.
- Knowledge of HTTP/HTTPS.
- Basic knowledge of Apache JMeter.
- Basic knowledge of e-commerce applications.

Theory / Concepts:

Performance testing is used to check the **speed, responsiveness, stability, and reliability** of an application under a particular workload.

In an e-commerce application, performance can be tested for:

- Login
- Product search
- Product viewing
- Add to cart
- Checkout

Important Metrics:

Response Time: Time taken by the application to respond to a request.

Throughput: Number of requests processed per second.

Error Rate: Percentage of failed requests.

Concurrent Users: Number of users accessing the application at the same time.

Advantages:

1. Identifies performance problems.
2. Improves application speed.
3. Checks the system under heavy load.
4. Improves user satisfaction.

Disadvantages:

1. Requires testing tools.
2. Testing can be time-consuming.
3. Requires additional resources.
4. Large-scale testing can be costly.

Algorithm:

1. Start.
2. Open Apache JMeter.
3. Create a new Test Plan.
4. Add a Thread Group.
5. Set the number of users and ramp-up time.
6. Add HTTP requests for login, product search, and cart.
7. Add a listener to view the results.
8. Run the test.
9. Measure response time and error rate.
10. Analyze the results.
11. Stop.

Application:

Performance testing is used in:

- E-commerce websites.
- Online shopping applications.
- Banking applications.
- Ticket booking systems.
- Food delivery applications.

Real-Life Example

During an online sale, thousands of users may access an e-commerce website simultaneously.

For example:

Login → Search Product → View Product → Add to Cart → Checkout

Performance testing checks whether the website can handle these users without becoming slow or unavailable.

Program / JMeter Configuration

Test Scenario

Tool : Apache JMeter

Number of Users : 100

Ramp-up Time : 20 seconds

Loop Count : 5

Test Plan

Test Plan

|

+-- Thread Group

|

+-- Login

+-- Search Product

+-- View Product

+-- Add to Cart

+-- Checkout

+-- Summary Report

HTTP Request

Method : GET

Path : /products

Login Request

Method : POST

Path : /login

username = testuser

password = test123

Use these values only with an authorized test application.

Sample Input

Number of Users = 100

Ramp-up Time = 20 seconds

Loop Count = 5

Operations:

1. Login
2. Search Product
3. View Product
4. Add to Cart
5. Checkout

Sample Output:

Total Requests = 2500

Successful Requests = 2475

Failed Requests = 25

Average Response Time = 850 ms

Maximum Response Time = 3200 ms

Throughput = 48 requests/sec

Error Rate = 1%

Output Interpretation

The application processed most requests successfully. The response time and error rate were measured to evaluate the performance of the application.

Viva Questions with Answers

Q1. What is performance testing?

Answer: Performance testing checks the speed, stability, and responsiveness of an application.

Q2. What is load testing?

Answer: Load testing checks the application under the expected number of users.

Q3. What is stress testing?

Answer: Stress testing checks the application beyond its normal load.

Q4. What is response time?

Answer: It is the time taken by an application to respond to a request.

Q5. What is throughput?

Answer: Throughput is the number of requests processed per second.

Q6. What is Apache JMeter?

Answer: Apache JMeter is an open-source tool used for performance and load testing.

Q7. What is a virtual user?

Answer: A virtual user is a simulated user created by a testing tool.

Q8. What is concurrent user?

Answer: A concurrent user is a user accessing the application at the same time as other users.

Q9. What is error rate?

Answer: Error rate is the percentage of failed requests.

Q10. What is scalability?

Answer: Scalability is the ability of an application to handle an increasing number of users.

Result:

The performance of the e-commerce application was successfully tested using Apache JMeter.

Exp.No:7	Automate the Testing of E-Commerce Applications Using Selenium
Date:	

Aim:

To automate the testing of an e-commerce application using Selenium WebDriver.

Objective:

- To automate e-commerce application test cases.
- To test login, product search, and cart operations.
- To reduce manual testing effort.
- To verify the functionality of the application.

Prerequisite Knowledge:

- Basic knowledge of software testing.
- Basic knowledge of HTML and web applications.
- Basic programming knowledge.
- Knowledge of Selenium WebDriver.
- Knowledge of Python.

Theory / Concepts:

Selenium is an open-source automation testing tool used to test web applications.

Selenium WebDriver controls a web browser automatically and performs actions such as clicking buttons, entering text, and verifying web-page results.

In an e-commerce application, Selenium can automate:

- Login
- Product search
- Product selection
- Add to cart
- Checkout

Advantages:

1. Reduces manual testing effort.
2. Saves testing time.
3. Provides accurate and repeatable results.
4. Supports multiple browsers.

Disadvantages:

1. Requires programming knowledge.
2. Initial test-script development takes time.
3. Difficult to maintain when the webpage changes.
4. Mainly used for web applications.

Algorithm:

1. Start.
2. Install Selenium WebDriver.
3. Open the browser.
4. Open the e-commerce application.
5. Enter username and password.
6. Click the Login button.
7. Search for a product.
8. Select the product.
9. Add the product to the cart.
10. Verify the cart.
11. Close the browser.
12. Stop.

Application:

Selenium automation can be used for:

- E-commerce websites.
- Online shopping applications.

- Banking websites.
- Booking websites.
- Educational websites.

Real-Life Example

An online shopping website can be tested automatically using Selenium.

The automated test performs:

Open Website → Login → Search Product → Select Product → Add to Cart → Verify Cart

This allows testers to execute the same test repeatedly without performing each step manually.

Program:

Python Selenium Program

```
from selenium import webdriver
```

```
from selenium.webdriver.common.by import By
```

```
import time
```

```
driver = webdriver.Chrome()
```

```
driver.get("https://www.saucedemo.com/")
```

```
driver.maximize_window()
```

```
# Login
```

```
driver.find_element(By.ID, "user-name").send_keys("standard_user")
```

```
driver.find_element(By.ID, "password").send_keys("secret_sauce")
```

```
driver.find_element(By.ID, "login-button").click()
```

```
time.sleep(2)
```

```
# Add product to cart
```

```
driver.find_element(By.ID, "add-to-cart-sauce-labs-backpack").click()
```

```
# Open cart
```

```
driver.find_element(By.CLASS_NAME, "shopping_cart_link").click()
```

```
time.sleep(2)
```

```
print("Product added to cart successfully")
```

```
driver.quit()
```

Sample Input:

URL : <https://www.saucedemo.com/>

Username : standard_user

Password : secret_sauce

Product : Sauce Labs Backpack

Sample Output:

Product added to cart successfully

The browser opens the website, logs in, adds the selected product to the cart, and verifies the cart operation.

Viva Questions with Answers

Q1. What is Selenium?

Answer: Selenium is an open-source tool used for automating web applications.

Q2. What is Selenium WebDriver?

Answer: WebDriver is a Selenium component that controls web browsers programmatically.

Q3. Which programming languages can be used with Selenium?

Answer: Java, Python, C#, JavaScript, and other supported languages.

Q4. What is find_element()?

Answer: It is used to locate an element on a web page.

Q5. What is click()?

Answer: It performs a click action on a web element.

Q6. What is send_keys()?

Answer: It is used to enter text into an input field.

Q7. What is WebDriver?

Answer: WebDriver is an interface that allows Selenium to communicate with and control browsers.

Q8. Can Selenium test desktop applications?

Answer: No. Selenium is mainly designed for web application automation.

Q9. What is quit()?

Answer: It closes the browser and ends the WebDriver session.

Q10. What is automation testing?

Answer: Automation testing uses software tools and scripts to execute test cases automatically.

Result:

The e-commerce application was successfully automated using Selenium WebDriver. The login and add-to-cart functionalities were tested successfully.

Exp.No:8	Integrate TestNG with Selenium Test Automation
Date:	

Aim:

To integrate TestNG with Selenium and automate the testing of an e-commerce application.

Objective:

- To use TestNG with Selenium WebDriver.
- To automate e-commerce test cases.
- To execute test cases using TestNG annotations.
- To generate test results automatically.

Prerequisite Knowledge:

- Basic knowledge of Java.
- Basic knowledge of Selenium WebDriver.
- Basic knowledge of TestNG.
- Basic knowledge of web applications.
- Basic knowledge of software testing.

Theory / Concepts:

TestNG is a testing framework for Java applications. It is commonly used with Selenium for organizing and executing automated test cases.

TestNG provides annotations such as:

- `@Test` – Defines a test method.
- `@BeforeMethod` – Executes before each test.
- `@AfterMethod` – Executes after each test.
- `@BeforeClass` – Executes before the test class.
- `@AfterClass` – Executes after the test class.

Using Selenium with TestNG makes test automation easier to organize and execute.

Advantages:

1. Easy to organize test cases.
2. Supports test annotations.
3. Generates test reports.
4. Supports multiple test cases.
5. Supports parallel execution.

Disadvantages:

1. Requires Java knowledge.
2. TestNG setup is required.
3. Test scripts need maintenance when the website changes.
4. Initial automation development takes time.

Algorithm:

1. Start.
2. Create a Java project.
3. Add Selenium and TestNG dependencies.
4. Create a test class.
5. Create a `@BeforeMethod` to open the browser.
6. Create `@Test` methods for e-commerce operations.
7. Perform login and add-to-cart operations.
8. Verify the expected result.
9. Create an `@AfterMethod` to close the browser.
10. Run the TestNG test.
11. View the test report.
12. Stop.

Application:

TestNG with Selenium can be used for:

- E-commerce websites.

- Banking websites.
- Online booking applications.
- Educational websites.
- Insurance websites.

Real-Life Example:

An e-commerce website can have multiple automated test cases:

Test 1: Verify Login

Test 2: Verify Product Selection

Test 3: Verify Add to Cart

TestNG can execute these test cases and provide a report showing whether each test **Passed** or **Failed**.

Program:

Java Selenium + TestNG Program

```
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;
import org.testng.Assert;
import org.testng.annotations.AfterMethod;
import org.testng.annotations.BeforeMethod;
import org.testng.annotations.Test;

public class EcommerceTest {

    WebDriver driver;

    @BeforeMethod

    public void setUp() {

        driver = new ChromeDriver();
```

```
driver.manage().window().maximize();  
driver.get("https://www.saucedemo.com/");  
}
```

@Test

```
public void loginTest() {
```

```
    driver.findElement(By.id("user-name"))  
        .sendKeys("standard_user");
```

```
    driver.findElement(By.id("password"))  
        .sendKeys("secret_sauce");
```

```
    driver.findElement(By.id("login-button"))  
        .click();
```

```
    String title = driver.getTitle();
```

```
    Assert.assertTrue(title.contains("Swag Labs"));
```

```
}
```

@Test

```
public void addToCartTest() {
```

```
    driver.findElement(By.id("user-name"))  
        .sendKeys("standard_user");
```

```
    driver.findElement(By.id("password"))  
        .sendKeys("secret_sauce");
```

```

        driver.findElement(By.id("login-button"))
            .click();

        driver.findElement(
            By.id("add-to-cart-sauce-labs-backpack"))
            .click();

        String cartCount = driver.findElement(
            By.className("shopping_cart_badge")).getText();

        Assert.assertEquals(cartCount, "1");
    }

    @AfterMethod
    public void tearDown() {
        driver.quit();
    }
}

```

Sample Input:

URL : <https://www.saucedemo.com/>

Username : standard_user

Password : secret_sauce

Test Cases:

1. Login Test
2. Add to Cart Test

Sample Output:

[TestNG] Running Tests

Login Test : PASSED

Add to Cart Test : PASSED

Tests Run : 2

Passed : 2

Failed : 0

Skipped : 0

Viva Questions with Answers

Q1. What is TestNG?

Answer: TestNG is a Java testing framework used to organize and execute test cases.

Q2. Why is TestNG used with Selenium?

Answer: TestNG helps organize Selenium test cases and provides test execution and reporting features.

Q3. What is @Test?

Answer: @Test identifies a method as a test case.

Q4. What is @BeforeMethod?

Answer: It executes before every test method.

Q5. What is @AfterMethod?

Answer: It executes after every test method.

Q6. What is Assert in TestNG?

Answer: Assert is used to compare the actual result with the expected result.

Q7. What happens when an assertion fails?

Answer: The corresponding test case is marked as failed.

Q8. What is a TestNG report?

Answer: It provides information about passed, failed, and skipped test cases.

Q9. Which language is used in this experiment?

Answer: Java.

Q10. Which tool is used for browser automation?

Answer: Selenium WebDriver.

Result:

The Selenium automation was successfully integrated with TestNG. The e-commerce login and add-to-cart test cases were executed successfully, and the TestNG results showed that the test cases passed.

Exp.No:9(a)	Build a data-driven framework using Selenium and Testing
Date:	

Aim:

To build a data-driven automation framework using Selenium WebDriver and TestNG for testing an e-commerce application with multiple sets of test data.

Objective:

- To separate test data from test scripts.
- To execute the same test with different input data.
- To use TestNG DataProvider.
- To automate e-commerce login testing.

Theory:

A **Data-Driven Framework** separates test data from the automation code. The same test case can be executed multiple times using different data.

TestNG provides the `@DataProvider` annotation for supplying multiple sets of data to a test method.

Advantages:

1. Same test can run with multiple data sets.
2. Reduces duplicate code.
3. Easy to maintain test data.
4. Improves test coverage.

Disadvantages:

1. Test data must be maintained properly.
2. Framework setup takes some time.
3. Large data sets can increase execution time.

Algorithm:

1. Create a Selenium and TestNG project.
2. Create a test class.

3. Create a @DataProvider.
4. Store multiple username and password values.
5. Pass the data to the test method.
6. Open the e-commerce website.
7. Enter the supplied login data.
8. Verify the login result.
9. Execute the test for all data sets.
10. Generate the TestNG report.

Program:

```
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;
import org.testng.Assert;
import org.testng.annotations.*;

public class DataDrivenTest {

    WebDriver driver;

    @BeforeMethod
    public void setUp() {
        driver = new ChromeDriver();
        driver.get("https://www.saucedemo.com/");
    }

    @DataProvider(name = "loginData")
    public Object[][] loginData() {
```

```
return new Object[][] {  
    {"standard_user", "secret_sauce"},  
    {"standard_user", "wrong_password"}  
};  
}
```

```
@Test(dataProvider = "loginData")
```

```
public void loginTest(String username, String password) {
```

```
    driver.findElement(By.id("user-name"))  
        .sendKeys(username);
```

```
    driver.findElement(By.id("password"))  
        .sendKeys(password);
```

```
    driver.findElement(By.id("login-button"))  
        .click();
```

```
    System.out.println("Login tested for: " + username);  
}
```

```
@AfterMethod
```

```
public void tearDown() {  
    driver.quit();  
}  
}
```

Sample Input:

Username	Password
----------	----------

standard_user	secret_sauce
---------------	--------------

standard_user	wrong_password
---------------	----------------

Sample Output:

Login tested for: standard_user

Login tested for: standard_user

Tests Run : 2

Passed : 2

Failed : 0

Results:

The Data-Driven Framework was successfully created using Selenium and TestNG, and multiple test data sets were executed successfully.

Exp.No:9(b) Date:	Build Page object Model using Selenium and TestNG
------------------------------------	--

Aim:

To build a Page Object Model framework using Selenium WebDriver and TestNG for e-commerce application testing.

Objective:

- To separate page elements from test cases.
- To improve code reusability.
- To make automation scripts easy to maintain.

Theory:

Page Object Model (POM) is a design pattern in Selenium where each web page is represented as a separate Java class.

The page class contains:

- Web elements.
- Locators.
- Methods for performing actions.

The test class contains the actual test cases.

Advantages:

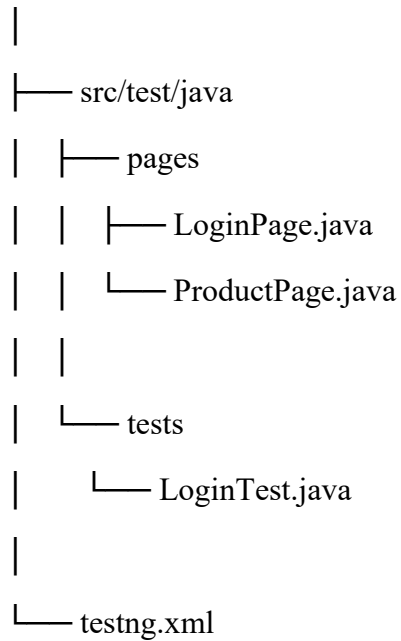
1. Reduces code duplication.
2. Easy to maintain.
3. Improves code readability.
4. Allows reuse of page methods.

Disadvantages:

1. Requires additional classes.
2. Initial development takes more time.
3. Changes in page elements require updates.

Framework Structure:

ECommerceProject



LoginPage.java

```
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;

public class LoginPage {

    WebDriver driver;

    By username = By.id("user-name");
    By password = By.id("password");
    By loginButton = By.id("login-button");

    public LoginPage(WebDriver driver) {
        this.driver = driver;
    }
}
```

```
public void enterUsername(String user) {  
    driver.findElement(username).sendKeys(user);  
}
```

```
public void enterPassword(String pass) {  
    driver.findElement(password).sendKeys(pass);  
}
```

```
public void clickLogin() {  
    driver.findElement(loginButton).click();  
}
```

```
public void login(String user, String pass) {  
    enterUsername(user);  
    enterPassword(pass);  
    clickLogin();  
}  
}
```

LoginTest.java

```
import org.openqa.selenium.WebDriver;  
import org.openqa.selenium.chrome.ChromeDriver;  
import org.testng.annotations.*;
```

```
public class LoginTest {
```

```
    WebDriver driver;
```

```
LoginPage loginPage;
```

```
@BeforeMethod
```

```
public void setUp() {
```

```
    driver = new ChromeDriver();
```

```
    driver.get("https://www.saucedemo.com/");
```

```
    loginPage = new LoginPage(driver);
```

```
}
```

```
@Test
```

```
public void verifyLogin() {
```

```
    loginPage.login("standard_user", "secret_sauce");
```

```
    System.out.println("Login successful");
```

```
}
```

```
@AfterMethod
```

```
public void tearDown() {
```

```
    driver.quit();
```

```
}
```

```
}
```

Sample Output:

Login successful

Tests Run : 1

Passed : 1

Failed : 0

Result:

The Page Object Model was successfully implemented using Selenium and TestNG, and the e-commerce login test was executed successfully.

Exp.No:9(c) Date:	Build BDD framework with Selenium, TestNG and Cucumber
------------------------------------	---

Aim:

To build a BDD framework using Selenium WebDriver, TestNG, and Cucumber for testing an e-commerce application.

Objective:

- To write test cases in simple English.
- To use Cucumber feature files.
- To integrate Cucumber with Selenium and TestNG.
- To improve communication between testers and developers.

Theory:

BDD (Behavior Driven Development) is a testing approach in which application behavior is described using simple, readable language.

Cucumber uses **Gherkin** syntax.

Important keywords are:

- Feature
- Scenario
- Given
- When
- Then
- And

Advantages:

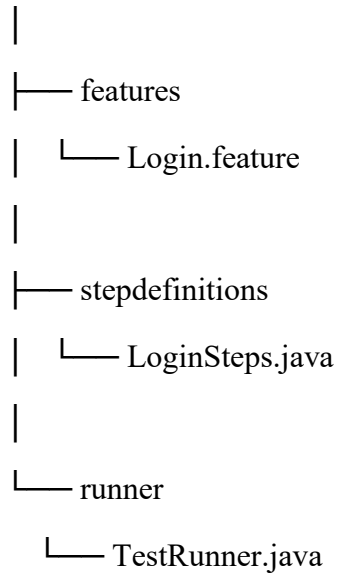
1. Test cases are easy to understand.
2. Supports readable scenarios.
3. Improves communication between team members.
4. Can be integrated with Selenium and TestNG.

Disadvantages:

1. Requires additional framework setup.
2. Feature files need maintenance.
3. Requires knowledge of Cucumber and Gherkin.

Framework Structure

ECommerceBDD



Feature File – Login.feature

Feature: E-Commerce Login

Scenario: Successful login

Given the user is on the e-commerce login page

When the user enters username "standard_user"

And the user enters password "secret_sauce"

And the user clicks the login button

Then the user should be logged in successfully

Step Definition – LoginSteps.java

```
import io.cucumber.java.en.*;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;

public class LoginSteps {
    WebDriver driver;

    @Given("the user is on the e-commerce login page")
    public void openLoginPage() {
        driver = new ChromeDriver();
        driver.get("https://www.saucedemo.com/");
    }

    @When("the user enters username {string}")
    public void enterUsername(String username) {
        driver.findElement(By.id("user-name"))
            .sendKeys(username);
    }

    @When("the user enters password {string}")
    public void enterPassword(String password) {
        driver.findElement(By.id("password"))
            .sendKeys(password);
    }

    @When("the user clicks the login button")
    public void clickLogin() {
        driver.findElement(By.id("login-button"))
            .click();
    }
}
```

```
@Then("the user should be logged in successfully")
public void verifyLogin() {
    System.out.println("Login successful");
    driver.quit();
}
}
```

TestNG Runner

```
import io.cucumber.testng.AbstractTestNGCucumberTests;
import io.cucumber.testng.CucumberOptions;
@CucumberOptions(
    features = "src/test/resources/features",
    glue = "stepdefinitions",
    plugin = {"pretty", "html:target/cucumber-report.html"}
)
public class TestRunner extends AbstractTestNGCucumberTests {
}
```

Sample Input

Username : standard_user

Password : secret_sauce

Sample Output

Feature: E-Commerce Login

Scenario: Successful login

Given the user is on the e-commerce login page PASSED

When the user enters username PASSED

And the user enters password PASSED

And the user clicks the login button PASSED

Then the user should be logged in successfully PASSED

1 Scenario Passed

5 Steps Passed

Viva Questions with Answers

Q1. What is a Data-Driven Framework?

Answer: It is a framework where test data is separated from the test script.

Q2. What is DataProvider in TestNG?

Answer: @DataProvider supplies multiple sets of data to a test method.

Q3. What is POM?

Answer: Page Object Model is a design pattern where each web page is represented by a separate class.

Q4. What is BDD?

Answer: BDD stands for Behavior Driven Development and describes application behavior in a readable format.

Q5. What is Cucumber?

Answer: Cucumber is a BDD testing tool that uses Gherkin language.

Q6. What is Gherkin?

Answer: Gherkin is the language used to write Cucumber feature files.

Q7. What does Given mean?

Answer: It describes the initial condition of the test.

Q8. What does When mean?

Answer: It describes an action performed by the user.

Q9. What does Then mean?

Answer: It describes the expected result.

Q10. Why is TestNG used with Selenium?

Answer: TestNG provides test organization, execution, assertions, and reporting features.

Result:

The BDD framework was successfully created using Selenium, TestNG, and Cucumber, and the e-commerce login scenario was executed successfully.